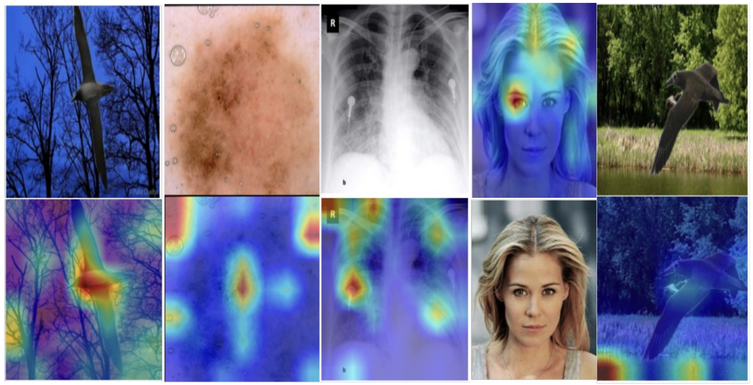
Label-Free Mitigation of Spurious Correlations in VLMs using Sparse Autoencoders

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Problem



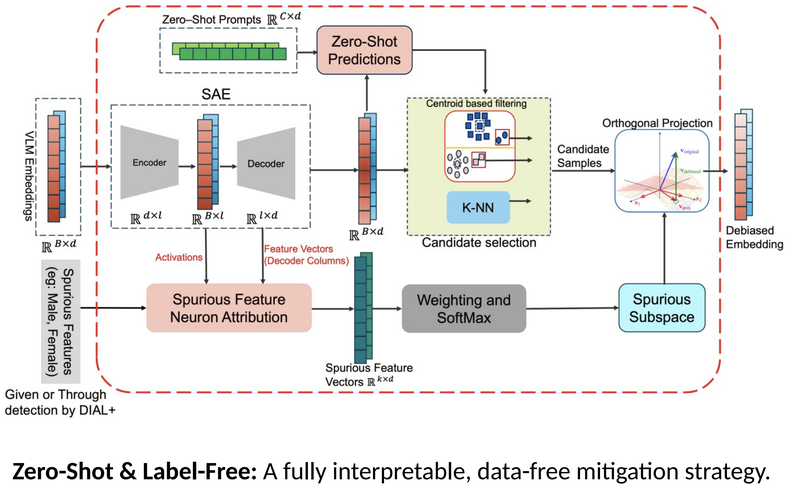
- Vision Language Models (VLMs) like CLIP are highly vulnerable to **spurious correlations** relying on coincidental artifacts rather than causal features.
- **Real-World Examples:** Models often rely on surgical skin markings rather than lesions to diagnose melanoma or focus on water backgrounds instead of the actual bird to classify waterbirds.
- **Impact:** Severe performance drops on specific semantic groups, causing robustness issues and compromising real-world fairness.

Prior Work:

- Current methods are **not interpretable** and require expensive workarounds: extra data, domain experts, or external LLMs.
- **High Resource Demands:** Fine-tuning and reweighting require expensive labeled datasets, domain expertise, or full model access.
- **Text-Only Focus:** Text based methods ignores the biases entrenched directly within the visual embeddings.

- [1] Lu, S., Chai, J., & Wang, X. (2025). Mitigating spurious correlations in zero-shot multimodal models. In *The Thirteenth International Conference on Learning Representations*.
- [2] Adila, D., Shin, C., Cai, L., & Sala, F. Zero-Shot Robustification of Zero-Shot Models. In *The Twelfth International Conference on Learning Representations*.

Method



- **Zero-Shot & Label-Free:** A fully interpretable, data-free mitigation strategy.
- **DIAL:** Mitigates known spurious biases directly in the VLM embedding space using **Sparse Autoencoders (SAEs).**
- **DIAL+:** Autonomously **detects and mitigates** unknown spurious biases.
- Matches or beats existing baselines across 5 major benchmarks.

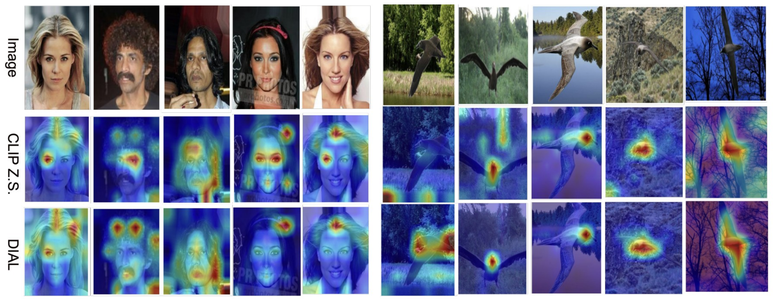
- Our framework debiases image embeddings in three rapid steps:
- **Candidate Selection:** Use zero-shot pseudo-labels to automatically flag samples that deviate significantly from their class centroids.
- **Disentanglement & Identification:** Pass embeddings through an SAE to unravel dense features into interpretable directions.

DIAL (Known Biases): Isolates specific feature directions that correlate with a provided spurious attribute

DIAL+ (Unknown Biases): Automatically detects hidden biases via simulated feature ablation, isolating "influential concepts" that flip predictions and are highly active in flagged candidate samples.

- **Orthogonal Mitigation:** Surgically subtract the "spurious subspace" from the flagged embeddings, preserving the causal data:

Results



CelebA | **Waterbirds**

Method	CLIP ViT-B/32			CLIP ViT-L/14			CLIP ViT-B/32			CLIP ViT-L/14		
	AVG (↑)	WG(↑)	Gap(↓)	AVG (↑)	WG(↑)	Gap(↓)	AVG (↑)	WG(↑)	Gap(↓)	AVG (↑)	WG(↑)	Gap(↓)
PerceptionCLIP	80.32	76.46	3.86	81.41	78.70	2.71	82.50	59.78	22.72	86.74	54.12	32.62
ROBOSHOT	84.77	80.52	4.25	85.54	82.61	2.93	71.92	54.41	17.51	64.43	45.17	19.26
TIE	85.11	82.63	2.48	86.17	84.60	1.57	79.82	71.35	8.47	84.12	78.82	5.30
TIE*	85.11	82.63	2.48	86.17	84.60	1.57	76.91	61.24	15.67	78.98	61.60	17.38
Zero-Shot	84.27	78.89	5.38	81.20	73.35	7.85	68.48	41.37	27.11	83.72	31.93	51.79
GroupPrompt	80.38	74.90	5.48	77.86	68.94	8.92	66.79	43.46	23.33	56.12	10.44	45.68
Ideal Words	80.96	78.12	2.84	**89.15**	76.67	12.48	**79.20**	**60.28**	**18.92**	**87.67**	64.17	23.50
Orth-Cali	82.31	77.92	4.39	81.39	77.69	3.70	69.19	54.99	14.20	86.31	58.56	27.75
DIAL (Ours)	**85.54**	**83.47**	2.17	86.87	**85.24**	1.63	71.88	52.82	19.06	82.6	68.69	13.91
DIAL+ (Ours)	85.28	83.42	**1.86**	86.54	85.15	**1.39**	68.48	42.26	26.22	82.25	**69.18**	**12.47**

Method	ISIC			COVID-19			FMOW		
	AVG (↑)	WG(↑)	Gap(↓)	AVG (↑)	WG(↑)	Gap(↓)	AVG (↑)	WG(↑)	Gap(↓)
PerceptionCLIP	52.74	41.55	11.19	56.87	48.84	8.03	17.70	12.61	5.09
ROBOSHOT	59.84	53.30	6.54	53.10	32.75	20.35	19.79	10.88	8.91
TIE	69.90	65.87	4.03	62.50	52.17	10.33	26.62	20.19	6.43
TIE*	71.68	61.11	10.57	61.08	50.22	10.86	26.65	19.84	6.81
Zero-Shot	70.21	42.21	28.00	**61.81**	44.83	16.98	26.02	18.06	7.96
GroupPrompt	30.05	12.13	17.92	48.27	27.58	20.69	14.69	8.75	**5.94**
Ideal Words	53.07	41.42	11.65	56.84	23.53	33.31	20.21	11.14	9.07
Orth-Cali	**72.54**	21.43	51.11	51.72	44.83	**6.89**	26.11	19.45	6.66
DIAL (Ours)	70.71	**68.42**	**2.29**	61.11	**48.28**	12.83	26.09	**19.90**	6.19
DIAL+ (Ours)	68.93	65.45	3.48	61.11	**48.28**	12.83	**26.23**	19.24	6.99

Debiased Retrieval on FairFace

Sensitive Feature	Original (MaxSkew@1000) (↓)	DIAL (MaxSkew@1000) (↓)
Age	1.32	**0.95**
Gender	0.30	**0.11**
Ethnicity	0.61	**0.32**

Conclusion

- **100% Zero-Shot & Label-Free:** DIAL and DIAL+ mitigate spurious correlations without auxiliary data, retraining, explicit labels, or LLMs..
- **Interpretable by Design:** SAEs and orthogonal projection offer transparent, verifiable debiasing that outperforms black-box methods and simple neuron ablation.
- **Future Directions:** A scalable foundation for machine unlearning and broader fairness applications in large multimodal models.